Fiscal Year:	FY 2010	**Task Last Updated:**	FY 07/02/2010
PI Name:	Pluth, Janice Ph.D.		
Project Title:	Elucidating the Relationship Between the Effects of Various Radiation Qualities and Cancer Development Processes using Novel Flow-based Assays		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
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PI Email:	janice.pluth@unlv.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	702-895-3236
Organization Name:	University of Nevada, Las Vegas		
PI Address 1:	Department of Health Physics and Diagnostic Sciences		
PI Address 2:	4505 S. Maryland Pkwy, Mail Code: 3037		
PI Web Page:			
City:	Las Vegas	**State:**	NV
Zip Code:	89154-9900	**Congressional District:**	1
Comments:	NOTE: Formerly at Lawrence Berkeley National Laboratory until fall 2017.		
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No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	Current radiation quality factors used for determining cancer risk for space radiation do not provide the accuracy needed, and there is debate as to the relevance of the experimental studies they're based upon. Therefore, one of NASA's primary goals is to improve the understanding of the carcinogenic process related to high LET exposures to better define the risk. Recently our current understanding of carcinogenesis has improved, and a number of cancer hallmarks have been identified. Some hallmarks, or surrogate cancer endpoints include genetic instability, telomere length changes and aberrant cellular signaling. Identifying relationships between early and late cell signaling profiles, later surrogate cancer endpoints, and radiation qualities will aid in determining a better risk assessment for cancer formation. In addition, using relevant cell types, such as human epithelial cells in which the majority of cancers develop is also key. In this proposal we will study critical phosphorylation signaling at early and late time points as well as surrogate cancer endpoints of		

Task Description:	telomere length changes and genomic stability as measured by centrosome amplification, at these same late times post exposure. Epithelial and fibroblast cells from the same individuals will be used to better understand the etiology of cancer, and why certain cell types more easily become carcinogenic. We will also perform studies to understand the heterogeneous nature of high LET radiation. Energy deposition from high LET ions is dense in the track cores, resulting in closely spaced and complex damages, and delta rays originating from cores provide additional exposures similar to X-rays. Using carefully chosen charge, energy and LETs of particles to obtain differing populations of cells hit by tracks and delta rays alone we will investigate how phosphorylation profiles change. These studies will allow us to better quantify the effects of space radiation and its relationship to the carcinogenic process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Brief description of scientific goals/objectives of the proposed research: In this proposal we are studying critical phosphorylation signaling at early (< 24 h) and late time points (days-weeks) following exposure. In addition, we will determine the relationship between these phospho-protein patterns, and two indicators of genomic stability and cancer progression, telomere shortening and centrosome amplification. Breast cancer studies have revealed that telomere length decreases during progression from usual ductal hyperplasia (UDH) to invasive cancer, thus telomere length changes can provide a readout of the carcinogenic process. A second marker we will be following after exposure is centrosome amplification which involves gains and losses of whole chromosomes and is believed to be due to continuous chromosome mis-segregation during mitosis resulting from centrosome defects. Combinations of increases in number, size, abnormal structure and function characterize centrosome amplification. Centrosome amplification has been detected in a broad range of solid tumors and more recently in leukemia and lymphoma. Centrosome abnormalities have been shown to increase with increasing tumor grade and aggressiveness. Over-expression of pericentrin, a centrosome protein, mimics many of the cellular characteristics of a high-grade carcinoma. Therefore centrosome abnormalities are also a very good indicator of cancer progression. Both of these potential markers for carcinogenesis will be followed using flow based assays. A difficulty with high-LET exposures, particularly at low doses, is that very uneven levels of damage are deposited within the population of cells. When studying effects at longer times post low dose exposure healthy cells will outgrow the population making it difficult to study effects in the minor population more severely affected. To study populations of cells exhibiting varying levels of damage we will isolate and individually expand populations separately for analysis of surrogate carcinogenic endpoints. We will utilize a reagent called CFSE which allows one to identify cells blocked at various cell cycles following damage, and thus isolate more heavily damaged cells based on cell cycle delays. Sets of mammary epithelial and fibroblast cells from the same individuals will be used to better understand the etiology of cancer, and whether certain cell types more frequently show evidence of genetic instability upon radiation exposure. These studies will complement our earlier studies using a normal hTERT immortalized skin fibroblast line, and help us identify unique signaling patterns specific to different cell types. By using two cell types from the same individual we will identify genotype influences in response. The relevance of considering d-rays when determining the risk of high-LET radiation will also be studied. Carefully chosen particles of various energies will be used to obtain differing populations of cells hit in a greater proportion by tracks or d-rays to investigate how phosphorylation profiles and carcinogenic endpoints are affected by these components. The intent of this proposal is to provide a comprehensive picture of HZE effects from initial phosphorylation patterns to later phosphorylation profiles, and the relationships to various surrogate markers for carcinogenesis. Studies will be performed using novel flow cytometry-based assays, which have been optimized in our lab, to more rapidly and less subjectively monitor these endpoints. Materials & Methods: Cells were set up in flasks or chamber slides prior to irradiation, irradiated at the various doses and then fixed at multiple time points thereafter. A methanol and a combination paraformaldehyde and methanol fixation were used. Fixed slides were sent back to our home institution in PBS to stain with appropriate primary and secondary antibodies. Cells in flasks were trypsinized off and fixed, then sent back on ice in methanol. Aliquots of fixed cells are then stained for various phospho-proteins and analyzed via flow cytometry. Brief summary of past years research accomplishments: During the first year efforts were concentrated on obtaining phospho-protein results in a normal fibroblast line, (82-6) following exposure to various combinations of ions selected to best identify the contribution of the d-rays versus the direct track biological effects. In the second year we have obtained replicate samples for this line as well as acquired results with a normal human primary mammary epithelial cell strain and an isogenic immortalized mammary fibroblast line. We also performed some preliminary studies in an isogenic breast mammary epithelial immortalized (thru chemical carcinogen exposure) line, to see if phospho-protein signaling is changed following radiation exposure if cells have an increased oncogenic potential. In comparing phospho-protein profiles for all cell strain and lines overall results indicate phosphorylation response is proportional to dose for all three phospho-proteins, although levels of induction differ between cell strain and lines. In comparing the mammary epithelial cell strain and line with the normal fibroblast cell line we see a much lower induction of gamma H2AX at all time points in the epithelial cells, with the immortalized line showing the lowest levels of induction. However at 24 h post exposure both the primary mammary epithelial line and the normal fibroblast line show equal levels of residual damages following high LET exposure and nearly complete repair following the low LET exposure. Thus although the different cell types appear to show differences in initial levels of induction, levels at longer times post exposure appear to be similar between normal lines with the immortalized line seeming to repair damages more rapidly. A larger comparison of phospho-profiles following numerous radiation quality exposures (Fe 300, Fe 600, Fe 1000, Si 173, Si 93 MeV/u and X-ray) in the normal fibroblast line and the primary mammary epithelial line reveal some general patterns. At early time points post exposure, gamma H2AX induction appears mainly dependent on LET, with Z2/ ß2 and energy playing some role as well. However the induction of pSMC1 appears primarily dependent on Z2/ ß2, and induction of pATF2 reflects a combination of of LET and Z2/ ß2 contributions. Overall this data suggests that far-reaching d-rays and energy deposited close to the track are the most critical in the induction of gamma H2AX and pATF2, whereas energy deposited in regions that are close and intermediate to the track produce the maximum biological effect when the endpoint assayed is ATM-mediated SMC1 phosphorylation. Thus radiation parameters such

	as LET, energy and d-rays contribute differently to cell signaling effects and the contribution of these parameters to biological response differs at various times post exposure. At 24 hr post exposure to all high LET radiation qualities, we observe residual levels of gamma H2AX reflecting the presence of unrepaired DSBs. In fibroblast cells the highest amount of residual damage and pATF2 phosphorylation is noted post exposure to Fe 600 MeV whereas similar levels of persistent phosphorylation of SMC1 is observed post all radiation qualities studied. A similar pattern is also observed in the 184D breast epithelial cells. High levels of residual ATF2 phosphorylation post exposure to Fe 600 MeV and Si 173 MeV indicate a significant contribution of LET and Z2/ ß2 to the late signaling effects post-exposure. These results would imply that ions depositing a greater density of d-rays contribute highest to damage induction and induction of pATF2, whereas close and intermediate energy depositions are of primary importance for both early and late effects of ATM-mediated SMC phosphorylation. Studies were also initiated analyzing surrogate cancer endpoints of telomere length and centrosome defects at many days to weeks post exposure in the same set of cell strains and lines. Results were exported from the flow cytometry files so that individual cells could be analyzed for overall cell telomere length changes and centrosome defects. Initial flow-based telomere studies reveal differences in radiation induced length changes dependent upon cell type, mammary epithelial cells appear to show a shortening of telomeres following radiation exposure which increases with dose, whereas fibroblast cells appear to show a lengthening of telomere that increases with dose. This latter effect has been observed before following radiation exposure and is believed to be due to chromosome healing, and/or upregulation of telomerase activity which can then add new telomeric repeats. It is interesting that these preliminary studies seem to suggest that mammary epithelial cells and mammary fibroblast cells appear to use different mechanisms of telomere repair (or non-repair in case of epithelial cells). Preliminary flow-based centrosome studies reveal the highest inductions occur in the low dose exposed samples (0.05 and 0.1 Gy), with two different fibroblast lines showing similar patterns of a peak in centrosome defects at the low dose region with a decrease with increasing dose and then a gradual increase again with further increasing doses. This seems to suggest that at the very low doses repair of centrosome defects does not occur and defects are likely duplicated with growth. This is of particular importance for astronauts who will be exposed to very low chronic doses during a lengthy mission.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Whalen MK, Sridharan, DM, Pluth JM. "Elucidating the Relationship Between the Effects of Various Radiation Qualities and Surrogate Cancer Endpoints Using Novel Flow-based Assays." EMS 2009: St. Louis – 40th Annual Meeting: Genomics in the Environmental Century, St. Louis, MO, October 24-28, 2009. Environmental and Molecular Mutagenesis, 2009. , Nov-2009
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